Individual Assignment 5

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Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrap axis labels
library(scales)

# reading .xlsx files
library(readxl)

# visualise missing data
library(naniar)

# arrange plots
library(patchwork)

# custom colour palette
library(NatParksPalettes)
```

Data

```r
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                    sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "sites")

# water quality
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))
```

```r
# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # filter out salinity outliers
  filter(date_site != "2022-08-12_NVBR")
```

```r
# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))
```

```r
# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         diptera_chironomid,
         annelida_polychaete) |>
  # filter to only include "filtered beaker" samples
  filter(sample_type %in% c("FB 250", "FB250")) |>
  # convert the data frame to long format
  pivot_longer(cols = ostracod:annelida_polychaete,
               names_to = "taxon",
               values_to = "abundance") |>
  # group by date, site, taxon
  group_by(date, site, taxon) |>
  # calculate average abundance per liter (sum abundance divided by 7.5 liters)
  summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
  # ungroup data frame
  ungroup() |>
  # create a new column called `date_site` to join with water quality
  unite("date_site", date, site, remove = FALSE) |>
  # join with water quality
  left_join(water_quality_clean, by = c("date_site")) |>
  # join with taxon list
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
  # add full names of sites
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "South of Venoco Bridge",
```

```r
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NWP" ~  "West Pond",
    site == "NDC" ~ "Devereux Creek"
  )) |>
  # setting factor levels for site
  mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm = TRUE),
         site_full = fct_rev(site_full))
```

1. Visualizing differences across sites in major taxon abundance

a. Plan your figure

x-axis: site_full y-axis: log_ave_lit geometry to represent average abundance/L: geom_jitter(height = 0, width = 0.1, shape = 21) geometry to represent medians: stat_summary(fun = median, color = "orange", size = 1)

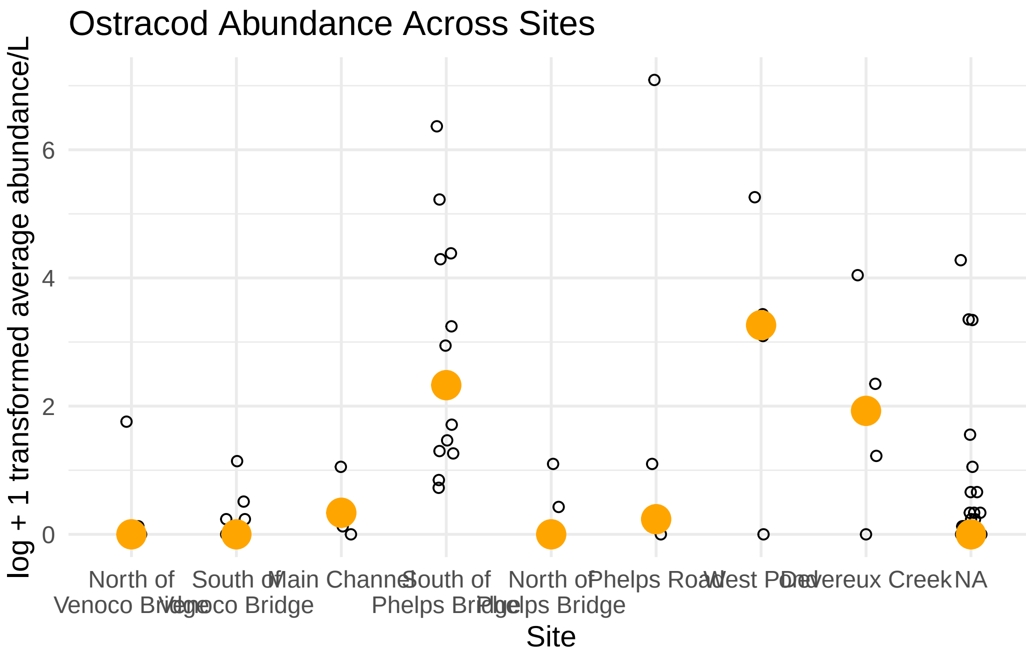
b. Wrangle your data

```r
# new object created from clean aquatic inverts
ostracods <- aq_ins_clean |>
  # filter to only include copepods
  filter(taxon == "ostracod") |>
  # log + 1 to transform mean abundance/liter
  mutate(log_ave_lit = log(ave_lit + 1))

slice_sample(ostracods)
```

```
# A tibble: 1 x 24
  date_site      date                site  taxon   ave_lit med_ph med_sal med_do
  <chr>          <dttm>              <chr> <chr>     <dbl>  <dbl>   <dbl>  <dbl>
1 2022-02-25_NEC 2022-02-25 00:00:00 NEC   ostrac~       0   8.75    44.1   9.87
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c. Code your figure

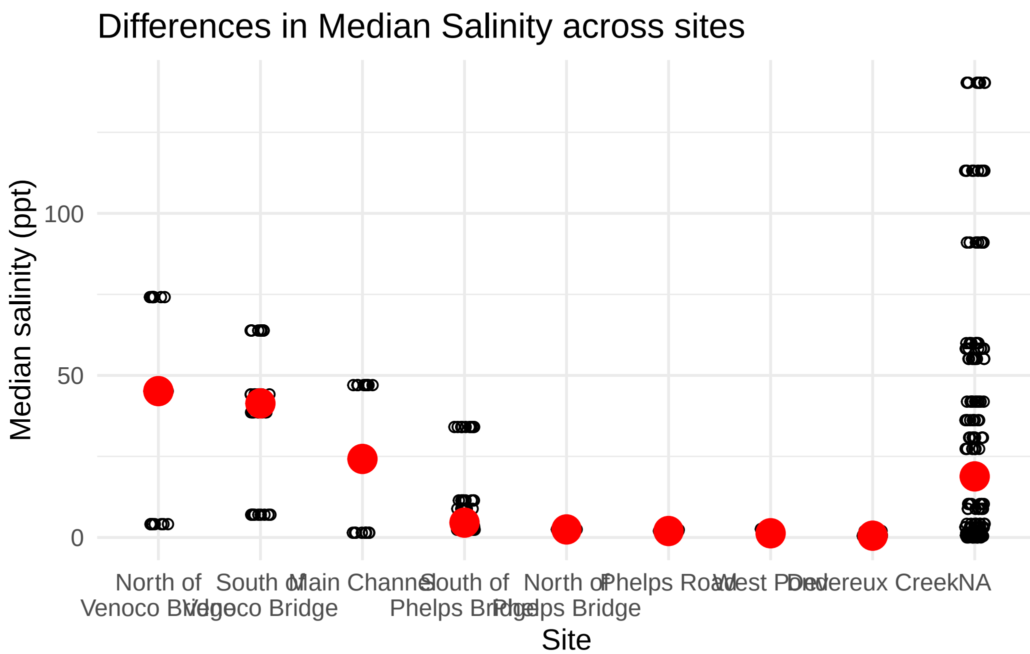
```
# base layer
ostracodgraph <- ggplot(data = ostracods,
       # x-axis: site
       # y-axis: mean salinity
       mapping = aes(x = site_full,
                     y = log_ave_lit)) +
  # salinity measurements at each survey
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(width = 15)) +
  # points to represent means
  stat_summary(fun = median,
               color = "orange",
               size = 1) +
  # labelling x-axis and y-axis
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Ostracod Abundance Across Sites") +
  # change theme from default
theme_minimal()

# displaying object
ostracodgraph
```



d. Create a multipanel figure

```
# salinity
salinity <- ggplot(data = aq_ins_clean,
       # x-axis: site
       # y-axis: mean salinity
       mapping = aes(x = site_full,
                     y = med_sal)) +
  # salinity measurements at each survey
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(width = 15)) +
  # points to represent means
  stat_summary(fun = median,
               color = "red",
               size = 1) +
  # labellingg x-axis and y-axis
  labs(x = "Site",
       y = "Median salinity (ppt)",
```

```
      title = "Differences in Median Salinity across sites") +
  # different theme
theme_minimal()
# displaying object
salinity
```

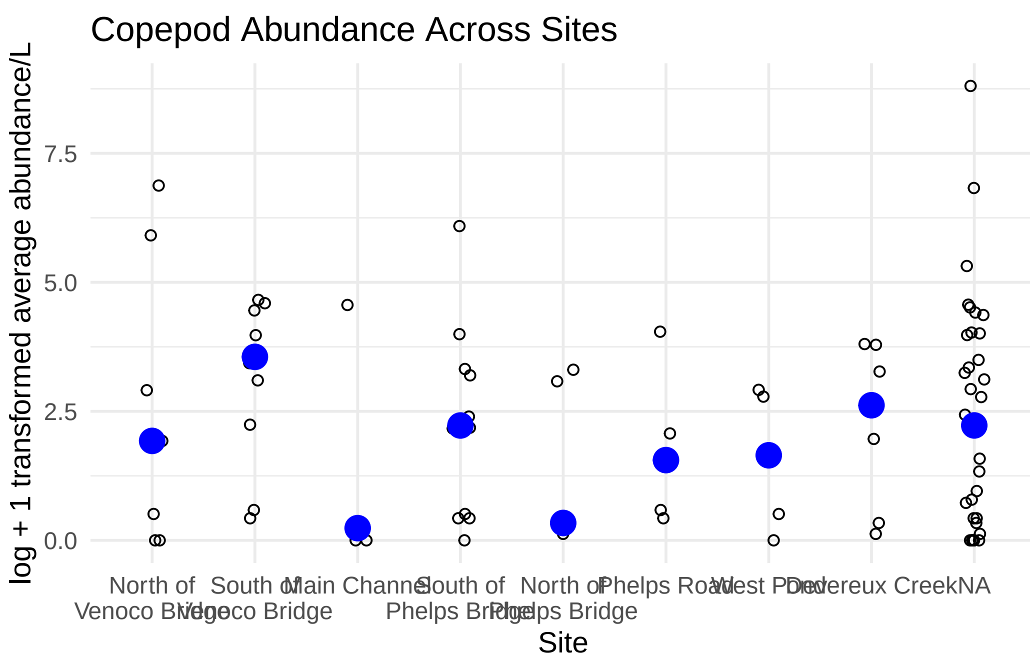



```
# cleaning copepods before graphing
copepods <- aq_ins_clean |>
  # filter to only include copepods
  filter(taxon == "copepod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))
8
```

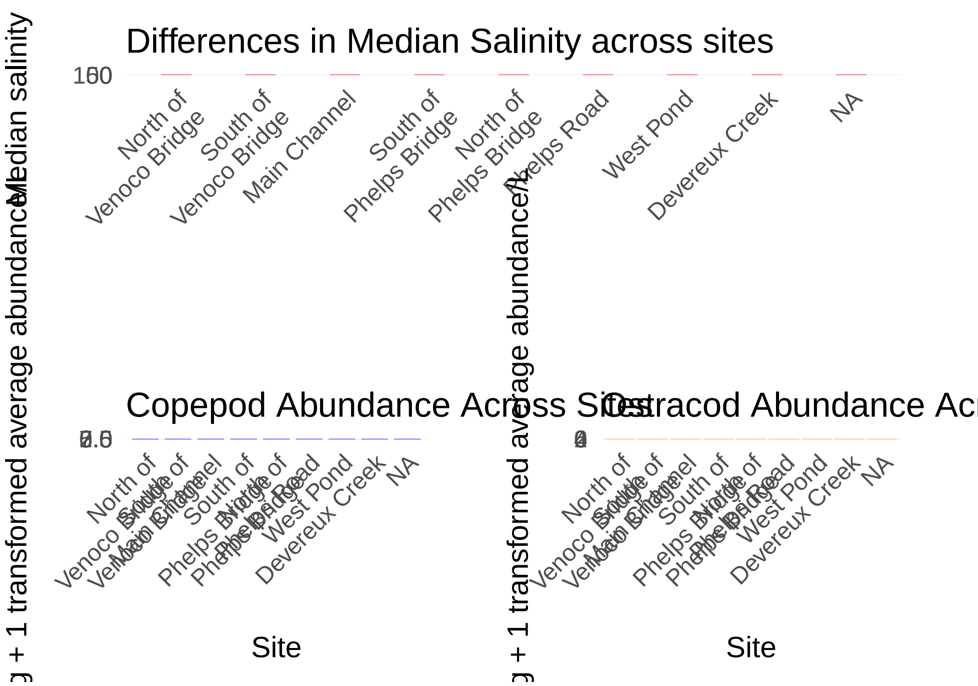
```
[1] 8
```

```
# base layer
copepods <- ggplot(data = copepods,
                   # x-axis: site
                   # y-axis: mean abundance/L
                   mapping = aes(x = site_full,
```

```
                                y = log_ave_lit)) +
  # first layer: points to represent observations
  geom_jitter(height = 0,
              shape = 21,
              width = 0.1) +
  # second layer: points to represent medians
  stat_summary(geom = "point",
               fun = "median",
size = 4,
               color = "blue") +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(15)) +
  # relabelling x- and y-axis, adding title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Copepod Abundance Across Sites") +
  # different theme
theme_minimal()
# displaying object
copepods
```



```r
combined_plot <-
  (salinity + labs(x = NULL)) /
  plot_spacer() /
  (copepods | ostracodgraph) +
  plot_layout(heights = c(1, 0.4, 1)) &
  theme(
    plot.margin = margin(15, 15, 15, 15),
    axis.text.x = element_text(angle = 45, hjust = 1),
    axis.title.x = element_text(margin = margin(t = 15)),
    axis.title.y = element_text(margin = margin(r = 15))
  )

combined_plot
```




```r
ggsave("multipanel_figure.png", combined_plot, width = 14, height = 10, dpi = 300)
```

e. Interpret your figure

- The Copepod and Ostracod abundance both vary across all sites. However, the sites where they are most and least abundant are different. For example, Copepods are most abundant at Venoco Bridge, Devereux Creek, and NA. However, Ostracods are most abundant at West Pond, South of Phelps Bridge, and Devereux Creek.

2. Visualizing total abundance as a function of salinity

a. Plan your figure

x-axis: Median salinity y-axis: Log transformed average abundance per liter color: scientificName geometry: geom_point() facets: facet_wrap(~scientificName, scales = "free")

b. Wrangle your data

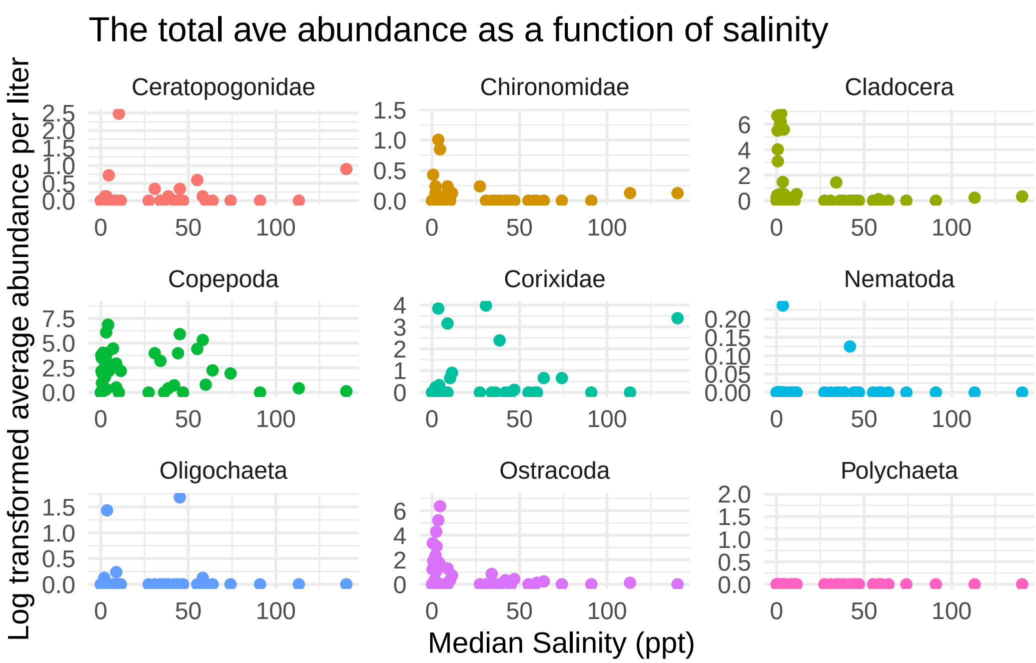
```
# new object created from aq_ins_clean
logsalinity2 <- aq_ins_clean |>
  # log + 1 transform mean abundance/liter
  mutate(log_ave_lit = log(ave_lit + 1))

slice_sample(logsalinity2)
```

```
# A tibble: 1 x 24
  date_site      date                site  taxon   ave_lit med_ph med_sal med_do
  <chr>          <dttm>              <chr> <chr>     <dbl>  <dbl>   <dbl>  <dbl>
1 2023-09-15_NMC 2023-09-15 00:00:00 NMC   copepod     0.4     NA      NA   14.2
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c. Code your figure

```
# plotting using  logsalinity2
logsal <- ggplot(data = logsalinity2,
       # x-axis: salinity
       # y-axis: log_ave_lit
       mapping = aes(x = med_sal,
                     y = log_ave_lit,
                     color = scientificName)) +
  geom_point() +
  # relabelling x-axis and y-axis and titling graph
  labs(x = "Median Salinity (ppt)",
       y = "Log transformed average abundance per liter",
```

```
        title = "The total ave abundance as a function of salinity") +
  # change theme from default
  theme_minimal() +
# creating multipanel figure
  facet_wrap(~scientificName, scales = "free") +
# removing legend
  theme(legend.position = "none")
# displaying plot
logsal
```



d. Interpret your figure

Majority of the taxa show their highest log-transformed average abundance/liter at lower salinities. This can be seen through many of the points clustered near the 0-10 ppt. Ostracoda and Cladocera both have the highest abundance values, but these points occur at lower salinities and are seen less at higher salinities. Additionally, Copepods appear across a wider range of salinities and still have a moderate abundance at mid-salinity sites. Furthermore, Nematoda, Oligochaeta and Polychaeta stay at a low abundance across most the salinity levels.